ASSIGNMENT 04 – PFP191

Duration: 70'

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Software Requirements

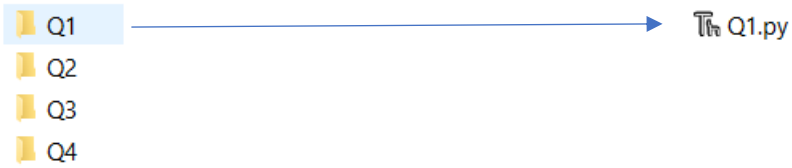
- Thonny, Notepad, Command Prompt, WinRAR / WinZip (or compress to ZIP file of Windows), Windows Explorer (File Explorer) on Windows 7 and above.

Students are ONLY Allowed to use:

- The materials like sample codes and program examples are stored on computer only.

Instructions

- Step 1: Students download the given materials from PEA Client.
- Step 2: Students read questions and prepare answers in the given template.
- Step 3: Submit a solution for each question:
    - The result is one folder for each question (Q1,Q2,Q3,Q4…), which contains **.py** source files.
      Example: Folder "Q1" contains Q1.py for question 1



    - For each question, you must submit one folder as above to PEA Client.

❖ **Importance:**

- Solutions will be marked by Automated Marking Software.
- **The use of tools other than those allowed in the above section whether intentionally or unintentionally, is considered a violation of the exam rules, and the mark is 0**
- Do not change the names of the folders, files, and, files and structure (format) of .py files specified in the exam. If you change it, the marking software can not find the execute file (.py) or the program output to mark, thus the score will be 0
- Do not edit given statements in the **main** function. If you change, the marking software can not mark and the score will be 0.

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Question 1: (2 marks)

The given file **Q1.py** already contains statements to input data for the collection. You should write statements to print the average of odd numbers in even places in the array. If none item, then print 0.00.

Notes:

- *Do not edit given statements in the **main** function*

- You can create new functions if you see it is necessary.

- The output result is formatted in two decimal places

Sample input and output:

Input: n = 6

Array: 1 3 7 4 9 11

After processing: (1+7+9) / 3 = 5.67

Output for marking:

OUTPUT:

5.67

Question 2: (3 marks)

The given file **Q2.py** already contains statements to input data for the collection. You should write statements to count the number of occurrences of the smallest elements in the array

Notes:

- *Do not edit given statements in the **main** function.*

- You can create new functions if you see it is necessary.

Sample input and output:

Input: n = 9

Array: 1 3 1 4 2 7 8 1 9

After processing: 3

Output for marking:

OUTPUT:

3

Question 3: (2 marks)

The given file **Q3.py** already contains statements to input data for the collection and an integer variable named **x**. You should write statements to print the average of all the elements are prime greater than x (the primes start from 2). If none prime greater than x then print 0.00.

Notes:

- *Do not edit given statements in the **main** function.*

- You can create new functions if you see it is necessary.

- The output result is formatted in two decimal places

Sample input and output:

Input: n = 9

Array: 1 3 5 4 2 7 8 11 9

Input: x = 3

After processing: (5 + 7 + 11) / 3 = 23/3 = 7.67

Output for marking:

OUTPUT:

7.67

Question 4: (3 marks)

The given file **Q4.py** already contains statements to input data for the collection. You should write statements to print the sum of the largest and smallest elements in the array.

Notes:

- *Do not edit given statements in the **main** function.*

- You can create new functions if you see it is necessary.

- The output result is formatted in two decimal places

Sample input and output:

Input: n = 10

Array: 2 3 5 4 2 7 8 11 9 11

After processing: (2 + 2 + 11 + 11) = 26

Output for marking:

OUTPUT:

26